A strictly positive averaged potential need not force global minimization

A counterexample to the spherical conjecture in arXiv:2104.03410

Candidate solution packet

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Status. Candidate full counterexample, likely valid pending expert review.

1 Source conjecture

For a continuous three-input kernel $K : (\mathbb{S}^{d-1})^3 \to \mathbb{R}$ and a probability measure μ, write

$$I_K(\mu) = \iiint K(x,y,z)\, d\mu(x) d\mu(y) d\mu(z), \qquad U_K^{\sigma}(x,y) = \int K(x,y,z)\, d\sigma(z),$$

where σ is normalized surface measure. Bilyk–Ferizović–Glazyrin– Matzke–Park–Vlasiuk conjecture, more generally for n inputs, that σ globally minimizes I_K if and only if the averaged two-input kernel $U_K^{\sigma^{n-2}}$ is conditionally positive definite [1]. Their Theorem 5.3 proves the necessary direction and proves only local minimality in the converse direction when the lower energy has σ as its unique minimizer.

The construction below disproves the global converse while satisfying the stronger uniqueness hypothesis in Theorem 5.3(ii).

2 Proof intuition

The averaged kernel U_K^{σ} records the quadratic variation of the three-input energy at σ. It cannot see a symmetric cubic interaction whose integral in one variable vanishes. Such an interaction can be locally flat at σ yet have a negative value at a concentrated measure.

We first build an explicit rotationally invariant cubic H_d with $U_{H_d}^{\sigma} = 0$ and negative Dirac energy. To rule out degeneracy as the explanation, we add a small symmetric lift of a centered exponential dot-product kernel. Its averaged kernel is strictly conditionally positive definite and uniquely minimized by σ. Taking the coefficient small preserves the negative Dirac gap. The result is a strict directional local minimum at σ which is not a global minimum.

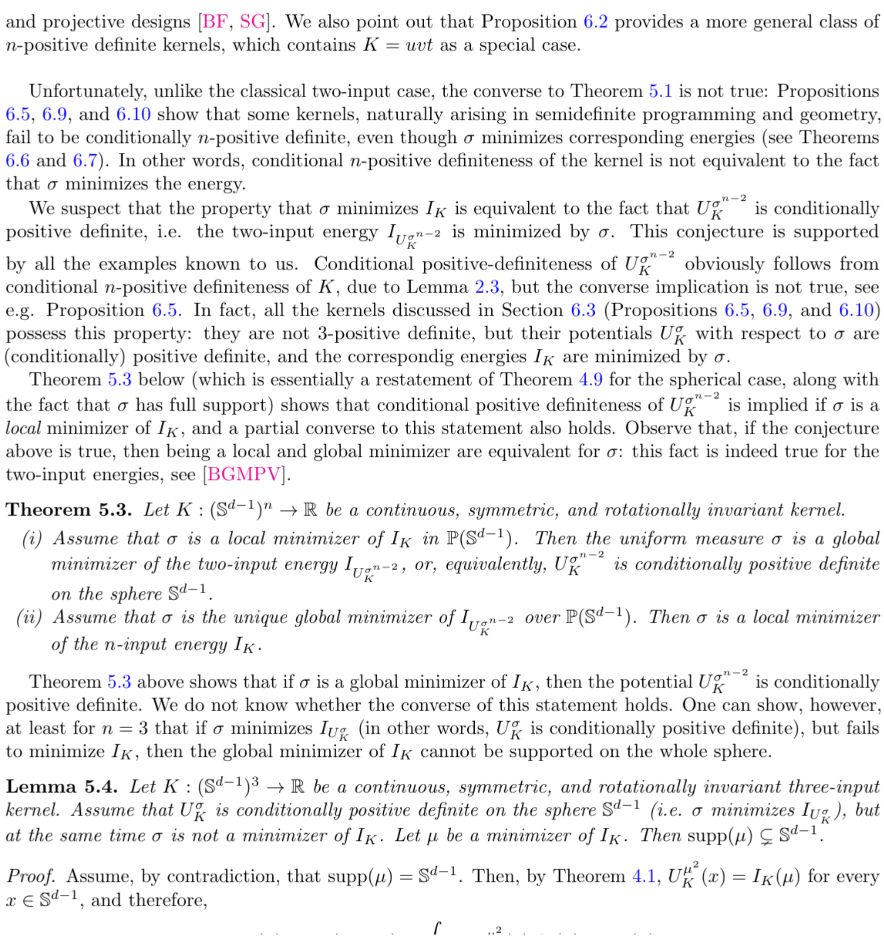
3 Counterexample

Fix $d \geq 3$. For $x, y, z \in \mathbb{S}^{d-1}$, set

$$u = \langle x, y \rangle, \qquad v = \langle y, z \rangle, \qquad t = \langle z, x \rangle.$$

Let

$$H_d(x,y,z) = -uvt + \frac{1}{d}(u^2 + v^2 + t^2) - \frac{2}{d^2}. \tag{1}$$


and projective designs [BF, SG]. We also point out that Proposition 6.2 provides a more general class of n-positive definite kernels, which contains $K = uvt$ as a special case.

Unfortunately, unlike the classical two-input case, the converse to Theorem 5.1 is not true: Propositions 6.5, 6.9, and 6.10 show that some kernels, naturally arising in semidefinite programming and geometry, fail to be conditionally n-positive definite, even though σ minimizes corresponding energies (see Theorems 6.6 and 6.7). In other words, conditional n-positive definiteness of the kernel is not equivalent to the fact that σ minimizes the energy.

We suspect that the property that σ minimizes I_K is equivalent to the fact that $U_K^{\sigma^{n-2}}$ is conditionally positive definite, i.e. the two-input energy $I_{U_K^{\sigma^{n-2}}}$ is minimized by σ. This conjecture is supported by all the examples known to us. Conditional positive-definiteness of $U_K^{\sigma^{n-2}}$ obviously follows from conditional n-positive definiteness of K, due to Lemma 2.3, but the converse implication is not true, see e.g. Proposition 6.5. In fact, all the kernels discussed in Section 6.3 (Propositions 6.5, 6.9, and 6.10) possess this property: they are not 3-positive definite, but their potentials U_K^σ with respect to σ are (conditionally) positive definite, and the correspondig energies I_K are minimized by σ.

Theorem 5.3 below (which is essentially a restatement of Theorem 4.9 for the spherical case, along with the fact that σ has full support) shows that conditional positive definiteness of $U_K^{\sigma^{n-2}}$ is implied if σ is a *local* minimizer of I_K, and a partial converse to this statement also holds. Observe that, if the conjecture above is true, then being a local and global minimizer are equivalent for σ: this fact is indeed true for the two-input energies, see [BGMPV].

Theorem 5.3. *Let $K : (\mathbb{S}^{d-1})^n \to \mathbb{R}$ be a continuous, symmetric, and rotationally invariant kernel.*
(i) Assume that σ is a local minimizer of I_K in $\mathbb{P}(\mathbb{S}^{d-1})$. Then the uniform measure σ is a global minimizer of the two-input energy $I_{U_K^{\sigma^{n-2}}}$, or, equivalently, $U_K^{\sigma^{n-2}}$ is conditionally positive definite on the sphere $\mathbb{S}^{d-1}$.
(ii) Assume that σ is the unique global minimizer of $I_{U_K^{\sigma^{n-2}}}$ over $\mathbb{P}(\mathbb{S}^{d-1})$. Then σ is a local minimizer of the n-input energy I_K.

Theorem 5.3 above shows that if σ is a global minimizer of I_K, then the potential $U_K^{\sigma^{n-2}}$ is conditionally positive definite. We do not know whether the converse of this statement holds. One can show, however, at least for $n = 3$ that if σ minimizes $I_{U_K^\sigma}$ (in other words, U_K^σ is conditionally positive definite), but fails to minimize I_K, then the global minimizer of I_K cannot be supported on the whole sphere.

Lemma 5.4. *Let $K : (\mathbb{S}^{d-1})^3 \to \mathbb{R}$ be a continuous, symmetric, and rotationally invariant three-input kernel. Assume that U_K^σ is conditionally positive definite on the sphere $\mathbb{S}^{d-1}$ (i.e. σ minimizes $I_{U_K^\sigma}$), but at the same time σ is not a minimizer of I_K. Let μ be a minimizer of I_K. Then $\mathrm{supp}(\mu) \subsetneq \mathbb{S}^{d-1}$.*

Proof. Assume, by contradiction, that $\mathrm{supp}(\mu) = \mathbb{S}^{d-1}$. Then, by Theorem 4.1, $U_K^{\mu^2}(x) = I_K(\mu)$ for every $x \in \mathbb{S}^{d-1}$, and therefore,

Figure 1: The conjecture, Theorem 5.3, and the repeated converse question on source PDF page 15.

Define the dimension-dependent constant

$$c_d = \int_{\mathbb{S}^{d-1}} e^{\langle p,z\rangle}\, d\sigma(z),$$

which is independent of $p \in \mathbb{S}^{d-1}$, and the centered pair kernel

$$G_d(x,y) = e^{\langle x,y\rangle} - c_d. \tag{2}$$

Since $\langle p,z\rangle < 1$ for σ-almost every z, one has $c_d < e$. Put

$$L_d(x,y,z) = G_d(x,y) + G_d(y,z) + G_d(z,x),$$

and choose

$$\eta_d = \frac{(d-1)(d-2)}{6d^2(e-c_d)} > 0, \qquad K_d = H_d + \eta_d L_d. \tag{3}$$

Theorem 1. *The kernel $K_d : (\mathbb{S}^{d-1})^3 \to \mathbb{R}$ is continuous, symmetric, and rotationally invariant. Its averaged pair kernel is*

$$U_{K_d}^\sigma = \eta_d G_d,$$

which is strictly conditionally positive definite; in particular, σ is the unique minimizer of its pair energy. The measure σ is a strict local minimizer of I_{K_d} in every direction, but it is not a *global minimizer. More precisely, for every $p \in \mathbb{S}^{d-1}$,*

$$I_{K_d}(\sigma) = 0, \qquad I_{K_d}(\delta_p) = -\frac{(d-1)(d-2)}{2d^2} < 0.$$

Consequently, the source conjecture and the converse question following Theorem 5.3 have negative answers.

Proof. Symmetry, continuity, and rotational invariance are immediate because a permutation of x, y, z merely permutes u, v, t, while simultaneous rotations preserve them.

Normalized surface measure satisfies

$$\int_{\mathbb{S}^{d-1}} zz^\mathsf{T}\, d\sigma(z) = \frac{1}{d} I_d.$$

For fixed x, y, this gives

$$\int vt\, d\sigma(z) = \frac{u}{d}, \qquad \int v^2\, d\sigma(z) = \int t^2\, d\sigma(z) = \frac{1}{d}.$$

Substitution in (1) yields

$$U^\sigma_{H_d}(x, y) = -\frac{u^2}{d} + \frac{1}{d}\left(u^2 + \frac{2}{d}\right) - \frac{2}{d^2} = 0. \tag{4}$$

By the definition of c_d, the σ-potential of G_d is zero. Hence

$$U^\sigma_{L_d}(x, y) = G_d(x, y), \qquad U^\sigma_{K_d} = \eta_d G_d.$$

We next prove strict conditional positive definiteness. If ν is a finite signed measure on $\mathbb{S}^{d-1}$ with $\nu(\mathbb{S}^{d-1}) = 0$, uniform convergence of the exponential series gives

$$\iint G_d(x, y)\, d\nu(x) d\nu(y) = \sum_{k=1}^{\infty} \frac{1}{k!} \left\| \int_{\mathbb{S}^{d-1}} x^{\otimes k}\, d\nu(x) \right\|^2 \geq 0. \tag{5}$$

If equality holds, every tensor moment of ν vanishes. Thus ν annihilates every polynomial restricted to $\mathbb{S}^{d-1}$. Those polynomials are dense in $C(\mathbb{S}^{d-1})$, so $\nu = 0$. This proves strictness.

For any $\mu \in \mathbb{P}(\mathbb{S}^{d-1})$, put $\nu = \mu - \sigma$. Since the σ-potential and σ-energy of G_d vanish,

$$I_{G_d}(\mu) = I_{G_d}(\nu) \geq 0,$$

with equality only for $\mu = \sigma$. Therefore σ is the unique minimizer of the pair energy of $U^\sigma_{K_d} = \eta_d G_d$.

Equations (4) and the zero potential of G_d imply $I_{K_d}(\sigma) = 0$. At a Dirac mass all three inner products are one, so

$$\begin{aligned} I_{H_d}(\delta_p) &= -1 + \frac{3}{d} - \frac{2}{d^2} = -\frac{(d-1)(d-2)}{d^2}, \\ I_{L_d}(\delta_p) &= 3(e - c_d). \end{aligned}$$

The choice (3) consequently gives

$$I_{K_d}(\delta_p) = -\frac{(d-1)(d-2)}{d^2} + \frac{(d-1)(d-2)}{2d^2} = -\frac{(d-1)(d-2)}{2d^2} < 0.$$

Thus σ is not a global minimizer.

For completeness, fix $\mu \neq \sigma$ and $\mu_s = (1-s)\sigma + s\mu$. Trilinearity, the vanishing one-point variation at σ, and $U^\sigma_{K_d} = \eta_d G_d$ give

$$I_{K_d}(\mu_s) = 3\eta_d s^2(1-s) I_{G_d}(\mu) + s^3 I_{K_d}(\mu). \tag{6}$$

The quadratic coefficient is strictly positive, so the right side is positive for all sufficiently small $s > 0$. Hence σ is a strict local minimizer in every direction, exactly in the source's sense. $\square$

4 Consequences and limitations

The example isolates the missing implication: even a strictly positive second variation at σ cannot control the cubic energy globally. It also shows that the word "unique" in Theorem 5.3(ii) is not enough to upgrade that theorem's local conclusion to a global one.

The source's necessary direction remains untouched: if σ globally minimizes the full energy, its averaged pair kernel is conditionally positive definite. Likewise, this example does not contradict the source's sufficient results assuming conditional three-positive-definiteness of the full kernel.

5 Verification and novelty status

The construction was independently checked through the spherical covariance identity, the tensor-series certificate (5), the exact Dirac energy, and the directional expansion (6). No numerical or external theorem is needed beyond density of polynomial restrictions on the sphere.

On August 11, 2026, the local run indexes were searched by arXiv id, exact title, the averaged-potential conjecture, and its local/global terminology. Exact-phrase and citation-oriented web searches found the source, its 2022 journal version, and related talks, but no later answer or matching counterexample. The search was bounded, so novelty confidence is moderate pending a specialist citation review.

References

[1] D. Bilyk, D. Ferizović, A. Glazyrin, R. W. Matzke, J. Park, and O. Vlasiuk, *Potential theory with multivariate kernels*, Math. Z. **301** (2022), no. 3, 2907–2935; arXiv:2104.03410; DOI: 10.1007/s00209-022-03000-z.